Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Baltic Sea Production Centre
BALTICSEA_MULTIYEAR_WAV_003_015

Issue: 2.0

Contributors: Hedi Kanarik, Laura Tuomi, Elisa Lindgren

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CHANGE RECORD

When the quality of the products changes, the QuID is updated, and a row is added to this table.

The third column specifies which sections or sub-sections have been updated. The fourth column should mention the reason for updating the document. Column 5 should include the names of the author(s) doing the changes. Including the name in column 6 shows that the document has been validated by the TAC/MFC PQ leader or deputy (or TAC/MFC PQ leader or the deputy). This validation means the document was checked for contents, formats and consistency with other documents.

Issue	Date	§	Description of Change	Author	Validated By
1.0	January 2020	All	First version of the QUID for the BAL MFC wave hindcast	Elisa Lindgren, Laura Tuomi, Vibeke Huess	Mercator Ocean
1.1	September 2023	1.1	Updated for operational temporal extension	V. Huess, Hedi Kanarik	Vibeke Huess
2.0	September 2024	All	Update of the entire product	Hedi Kanarik, Laura Tuomi	Vibeke Huess

ACRONYM TABLE

Acronym	Description
TAC	Copernicus Marine Service Thematic Assembly Centre
MFC	Copernicus Marine Service Monitoring Forecasting Centre
SWH	Significant Wave Height
WTPK	Peak Wave Period
MWT	Mean Wave Period

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This Quality Information Document (QUID) covers the BALTICSEA_MULTIYEAR_WAV_003_015 product produced by the Copernicus Marine Environment Monitoring Service (CMEMS) Baltic Sea Monitoring and Forecasting Centre and aims to qualify the product through a comprehensive validation against observational data. The product covers the period 01/01/1980 to around four months before present date. The product will be extended monthly. The validation covers the period of 1980-2023 from the areas for which there are in-situ and satellite wave data available. Extensions to the product undergo internal validation to ensure a similar quality to the one presented in this document. The parameters validated include significant wave height (SWH), mean wave period (MWT) and peak wave period (WTPK).

The time series use the model grid points that are closest to the observations. The resulting gridded data have a horizontal resolution of approx. 1 nautical mile (nmi) (i.e., 1’ latitude and 1’ 40’’ longitude).

This QUID-report describes the quality of the CMEMS Baltic Sea wave product as made available in November 2024.

I.2 Summary of the results

The quality of the Baltic Sea wave hindcast has been assessed by using observational data from 24 wave buoys for the period 1980 to 2023 and 12 satellite altimeters since 2002. Wave buoy measurements are divided in open sea and near-coastal measurements, the latter defined as those with a maximum distance of 15 km from the coast.

Measurements used for validation were chosen to represent different areas evenly and to cover the whole modelled period from the 1980’s. For example, some measurements were discarded from the analysis if there were overlapping periods of measurements from buoys near each other. Seasonal ice cover shortens the annual measurement period in the northernmost part of the Baltic Sea. The number of match-up pairs is lower at these buoy sites compared to sites in the southern Baltic Sea that were measured over the same period.

The main results for each of the parameters are as follows:

Significant wave height:

Significant wave height (SWH) is fairly well represented by the model (see Table 1, Table 2 and Table 3) with zero bias in both open sea and nearshore areas. Comparison against satellite altimeters however show a slight negative bias (-0.02 m). The root mean square difference (RMSD) varied between 0.21 and 0.24 m and overall correlation was between 0.95-0.96.

Wave period at spectral peak:

The results for wave period at spectral peak (WTPK) show an accuracy typical of wave models in the Baltic Sea (e.g., Tuomi 2008). The peak periods related to higher values of significant wave height are fairly well reproduced by the model but there is more scatter among peak periods related to low significant wave heights. The model performs better in the estimation of peak period at the open sea areas than near-coastal areas. Open sea stations having bias of -0.22 s and coastal stations -0.29 s (Table

1 and Table 2). The scatter of the values is significantly lower at open sea areas with 1.00 s RMSD and correlation of 0.70 compared to coastal RMSD of 1.53 s and correlation of 0.78.

Mean wave period:

The mean wave period (MWT) shows a similar accuracy as wave period at spectral peak (Table 1 and Table 2). The mean wave period is slightly underestimated by the model with better accuracy at the open sea areas. At the open sea, bias is –0.68 s and RMSD 0.81 s compared to coastal bias of –0.77 s and RMSD of 0.94 s. Correlation was however of the same order of magnitude with only a slightly higher value at the open sea (0.87) compared to coastal sites (0.86).

Other parameters:

All the available parameters in the wave forecasts are calculated from the modelled wave spectrum. While not all parameters have been validated, the accuracy of these parameters is estimated to be of the same order of magnitude.

I.3 Estimated Accuracy Numbers

Table 1 *Estimated accuracy numbers for significant wave height (SWH), peak wave period (WTPK) and mean wave period (MWT). Estimates are based on a comparison with open sea wave buoy data. See list of the sites from* ***Table 4*** *and Figure2. Equations for bias, mean absolute deviation (MAD), root mean-square difference (RMSD) and correlation are defined in Section III.1.*

	Bias	MAD	RMSD	Correlation	unit
SWH	0.00	0.15	0.21	0.95	m
WTPK	-0.22	0.65	1.00	0.78	s
MWT	-0.68	0.69	0.81	0.87	s

Table 2 *Estimated accuracy numbers for significant wave height (SWH), peak wave period (WTPK) and mean wave period (MWT) from coastal wave buoys. Estimates are based on a comparison with wave buoy data that are closer than 15 km from the coast. See list of the sites from* ***Table 5*** *and Figure 2. Equations for bias, mean absolute deviation (MAD), root mean-square difference (RMSD) and correlation are defined in Section III.1.*

	Bias	MAD	RMSD	Correlation	unit
SWH	0.00	0.16	0.22	0.96	m
WTPK	-0.29	0.95	1.53	0.70	s
MWT	-0.77	0.79	0.94	0.86	s

Table 3 *Estimated accuracy numbers for significant wave height on main altimeter frequency band (VHM0) validated over the period of 2002-2023. See list of the altimeters used from* ***Table 6****. Equations for bias, mean absolute deviation (MAD), root mean-square difference (RMSD), correlation and scatter index (SI) are defined in Section III.1.*

	Bias	MAD	RMSD	Correlation	unit
SWH	-0.02	0.18	0.24	0.95	m

II PRODUCTION SYSTEM DESCRIPTION

The product BALTICSEA_MULTIYEAR_WAV_003_015 is based on simulations with the WAM cycle 4.7 (e.g. Komen et al. 1994) using the ST4 source term package (Ardhuin et al. 2010). This product includes some modifications and additions to account for specific features of the Baltic Sea (Tuomi et al., 2011 and 2014).

Differences compared to the operational forecast product BALTICSEA_ANALYSIS_FORECAST_WAV_003_010

- The wind-wave coupling Betamax value has been tuned to match the quality of the forcing ERA5 wind fields in the Baltic Sea, which underestimate wind speed especially during high wind events.
- The hindcast does not include wave-current interactions nor sea surface height variation.
- Seasonal ice cover is taken into account using a threshold of 30 % above which wave model grid points are excluded from the calculation. Due to the coarse resolution of the ice data, additional grid obstructions are not used. Ice concentration values are taken from ice charts, and thus this information updates only once per day compared to hourly ice concentrations in forecast product. Daily ice charts are available since 2005; before this, the update frequency is 3 to 4 days.
- The directional resolution of the model wave spectrum is 15 degrees, whereas in the NRT system it is 10 degrees.

Production

BALTICSEA_MULTIYEAR_WAV_003_015 is produced at FMI (Finnish Meteorological Institute), Finland.

Summary of the model setup

Name of the forecast model: WAM cycle 4.7

Model domain: Baltic Sea

Lateral product boundaries: 53°N - 66°N, 9°E - 30°E with 1 nautical mile horizontal resolution

Wave model and source term physics configuration:

Wave model WAM solves the spectral action balance equation (Komen et al. 1994), which can be written as:

$$\frac{\partial}{\partial t}\left(\frac{F}{\sigma}\right) + c_g \cdot \frac{\partial}{\partial x}\left(\frac{F}{\sigma}\right) - (\nabla_x \sigma) \cdot \frac{\partial}{\partial k}\left(\frac{F}{\sigma}\right) = \frac{S_{tot}}{\sigma}$$

where $F(k, x, t)$ is the wave energy spectrum, k is the wave number, h is the water depth, $\sigma(k, h)$ is the intrinsic frequency, c_g is the group velocity and S_{tot} is the sum of the source terms. The source terms include the wind input and dissipation of waves due to whitecapping (Ardhuin et al., 2010) discrete approximation of weakly non-linear wave-wave interactions (Hasselmann et al. 1985) and wave dissipation due to depth-induced wave breaking (Battjes and Janssen, 1978) and bottom friction (Hasselmann et al. 1973).

The WAM cycle 4.7 accounts for unresolved islands within the given grid resolution using a method originally proposed by Tolman (2003). The land-sea mask and obstruction grids for the archipelago areas of the northern Baltic Sea have been prepared using the method presented in Tuomi et al. (2014). The

model uses meteorological forcing from ERA5 reanalysis, boundary spectra from ERA5 wave reanalysis at the open boundary at Skagerrak, and ice concentrations from SMHI/FMI's gridded ice charts.

Spectral grid: The two-dimensional frequency-direction grid for the wave spectra is comprised of 35 logarithmically spaced frequencies (0.0418 - 1.067 Hz) and 24 directions using 15-degree intervals.

Atmospheric forcing: ERA5 reanalysis for wind, resolution 0.28 degrees (approximately 31 km).

Boundary forcing: Boundary spectra from ERA5 wave reanalysis at the open boundary at Skagerrak.

Ice conditions: Ice concentration from SMHI/FMI's gridded ice charts. The resolution of the digitized product is coarser for the time period 1980-2015, 3 nmi, and 2015-2017, 2.4 nmi and higher, 0.25 nmi for 2017 onward, periods (Figure1)

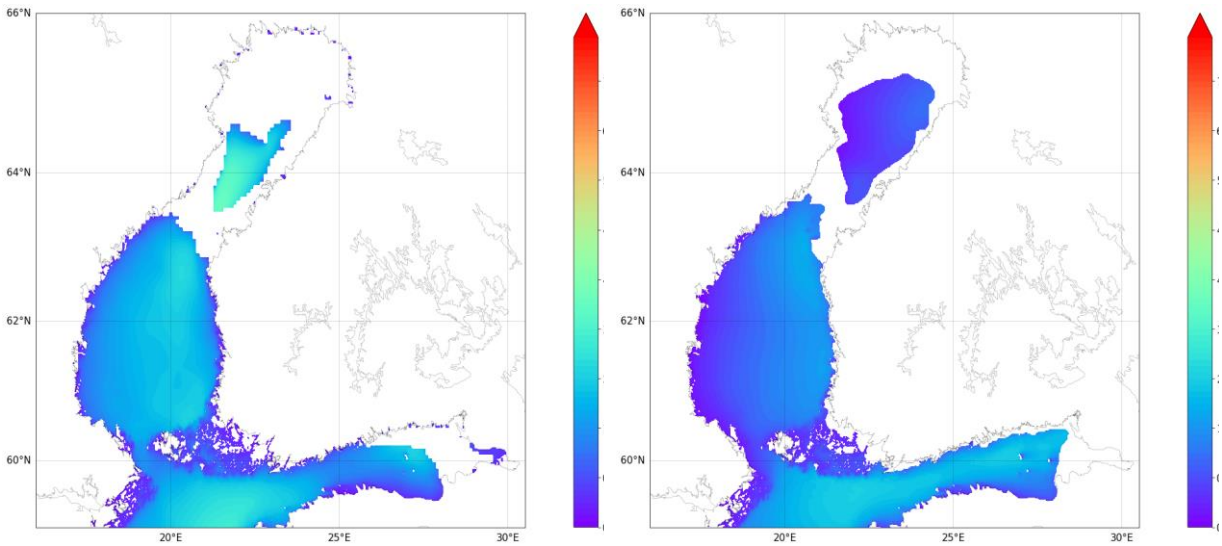

Figure 1 *Differences in ice field resolution. Example ice fields in January 1998 with a resolution of 3 nautical miles (nmi, left) and January 2018 with a resolution of 0.25 nautical miles (right).*

III VALIDATION FRAMEWORK

III.1 Validation metrics

This quality assessment report includes results from a 1980-2023 qualification period performed at the Finnish Meteorological Institute (FMI). The buoy data available from each buoy site is summarized in Table 4 and Table 5 and availability of different satellite data in Table 6. The hindcast simulations are saved with hourly resolution.

These core target products have been rated for the product BALTICSEA_MULTIYEAR_WAV_003_015:

SWH/VHM0: Spectral significant wave height (Hm0) [m]

WTPK/VTPK: Wave period at spectral peak (Tp) [s]

MWT/VTM02: Mean wave period of spectral moments (0,2), Tm02 [s]

SWH is validated for all wave measurement sites (locations shown in Figure2) and from each satellite. WTPK and MWT are validated from buoy observation using whichever parameters were available from each measurement site (listed in Table 4 and Table 5).

The following metrics were used to evaluate statistical differences between the BALMFC wave product and reference dataset:

1. number of data values used to compute the statistic
2. mean of product
3. mean of reference
4. standard deviation of product
5. standard deviation of reference
6. Bias, mean absolute deviation (MAD), root mean-square difference (RMSD), Pearson Correlation and Scatter Index (SI)
7. Quantile-quantile (qq) comparison of model versus observation distributions
8. Scatter diagrams and linear fit relationships of model versus observations

Formulae used for the calculation are:

$$mean(x) = \frac{1}{N}\sum_{i=1}^{N} x_i$$

$$var(x) = \frac{1}{N}\sum_{i=1}^{N}(x_i - mean(x))^2$$

$$std(x) = \sqrt{(var(x))}$$

$$cov(x,y) = \frac{1}{N}\sum_{i=1}^{N}(x_i - mean(x))(y_i - mean(y))$$

$$MSE(x,y) = \frac{1}{N}\sum_{i=1}^{N}(x_i - y_i)^2$$

$$RMSE(x,y) = \sqrt{MSE(x,y)}$$

$$CORR(x,y) = \frac{cov(x,y)}{\sqrt{var(x)var(y)}}$$

$$BIAS(x,y) = (mean(x) - mean(y))$$

$$MAE(x,y) = \frac{1}{N}\sum_{i=1}^{N}|x_i - y_i|$$

$$SI(x,y) = \frac{RMSE(x,y)}{mean(y)}$$

III.2 Wave in-situ measurement sites

A total of 24 measurement sites were used for validation of the BALMFC wave product (Figure2). Only measurement sites that had more than two months of quality assured data for the qualification period were selected (CMEMS product INSITU_BAL_NRT_OBSERVATIONS_013_032). The buoy locations, parameters used in the validation, and the period of data available are listed in Table 4 and Table 5. Buoy locations were divided into open sea areas, typically used for the validation of the Baltic Sea wave products (Table 4) and coastal wave measurements that have distance maximum 15 km from coast (Table 5). This was done to evaluate model performance also near coast even though some coastal stations have shorter measurement campaigns compared to many open sea measurement stations. Helsinki buoy (Figure2) makes an exception of this rule and is defined as open sea buoy as it is the central most measurement site at the Gulf of Finland. Some measurement stations were excluded from comparison to ensure even temporal and spatial distribution of measurement sites. For example, from two buoys close to each other with overlapping temporal coverage, only one was chosen for comparison.

III.3 Wave altimeter measurements

Observations from 12 different satellites (Table 6) were utilized in the validation of the Baltic Sea wave products (CMEMS products: WAVE_GLO_PHY_SWH_L3_MY_014_005 and WAVE_ GLO_PHY_SWH_L3_NRT_014_001). Time periods with data available over the Baltic Sea for each satellite are marked in the Table 6. Only significant wave was validated against altimeters.

Table 4 *Baltic Sea wave measurement sites used in the validation of the product BALTICSEA_MULTIYEAR_WAV_003_015. VTZA stands for average zero crossing wave period.*

STATION	LATITUDE	LONGITUDE	PARAMETERS	TIME PERIOD
DarsserSWR	54.6870	12.6890	VHM0, VTPK, VTZA	1991-2001
HuvudskarOst	58.9333	19.1667	VHM0, VTPK, VTZA	2001-2022
Northern Baltic	59.2500	21.0000	VHM0, VTPK	1996-2023
Helsinki Buoy	59.9650	25.2350	VHM0, VTPK	1990-1994, 2000-2023
FinngrundetWR	60.9000	18.6167	VHM0, VTPK, VTZA	2006-2023
Bothnian Sea	61.8000	20.2333	VHM0, VTPK	1998-1999, 2017-2023
Bothnian Bay	64.6842	23.2392	VHM0, VTPK	2017-2023
Knollsgrund	57.5227	17.5973	VHM0, VTPK, VTZA	2011-2022
Sodra Ostersjon	55.9200	18.7978	VHM0, VTZA	2006-2011
OlandsSodraGrund	56.0667	16.6833	VHM0, VTPK, VTZA	1980-2004
Ostergarn	57.4333	19.2833	VHM0, VTZA	1986
LasoOst	57.1654	11.5822	VHM0, VTZA	2001-2009

Table 5 *Coastal Baltic Sea wave measurement sites used in the additional validation of the product BALTICSEA_MULTIYEAR_WAV_003_015. VTZA stands for average zero crossing wave period*

STATION	LATITUDE	LONGITUDE	PARAMETERS	TIME PERIOD
Almagrundet	59.1500	19.1333	VHM0, VTZA	1980-1988
Trubaduren	57.6000	11.6333	VHM0, VTZA	1980-2003
Ulkokalla	64.2683	23.3583	VHM0, VTPK	1980-1981
Hoburg1	56.8333	18.2167	VHM0, VTZA	1981-1982
GustavDahlen	58.6000	17.4667	VHM0, VTZA	1983-1987
Laholmsbukten	56.5500	12.7333	VHM0, VTZA	1984-1985
Gotland	57.4167	19.0533	VHM0, VTPK	1995-2021
VaderoarnaWR	58.4833	10.9333	VHM0, VTPK, VTZA	2005-2023
VilsandiOuter	58.4892	21.6325	VHM0, VTZA	2015-2016
Suomenlinna	60.1233	24.9725	VHM0, VTPK	2017-2023
RaumaOutter	61.1999	21.0834	VHM0, VTPK	2020
IsokariEast	60.6846	20.5839	VHM0, VTPK	2023

Table 6 *Satellite products used in the validation of the product BALTICSEA_MULTIYEAR_ WAV_003_015. VAVH corresponds to significant wave height on main altimeter frequency band.*

Satellite / Altimeter	PARAMETERS	TIME PERIOD
Envisat	VAVH	2002-2012
Jason-1	VAVH	2002-2012
Jason-2	VAVH	2008-2017
Jason-3	VAVH	2016-2019, 2021-2023
SARAL/AltiKa	VAVH	2013-2019, 2021-2023
Sentinel-3A	VAVH	2021-2023
Sentinel-3B	VAVH	2021-2023
Sentinel-6A	VAVH	2021-2023
HaiYang-2B	VAVH	2021-2023
HaiYang-2C	VAVH	2022-2023
Cryosat-2	VAVH	2010-2015, 2021-2023
SWOT nadir	VAVH	2023

Figure 2 *Baltic Sea wave measurement sites used in the validation of the BALMFC wave product. Red diamonds indicate more open sea wave buoys listed in Table 4 whereas orange squares show more coastal wave buoys listed in Table 5.*

IV VALIDATION RESULTS

IV.1 Significant wave height

The hindcast significant wave height was compared to observations from 24 Baltic Sea measurement sites and 12 satellite altimeters. The statistical quantities for each measurement site are presented in Table 7 and for each altimeter in Table 8. The scatter-quantile-quantile (qq) plot in Figure3 shows the wave model quality over the whole domain against open sea (upper left) and coastal (upper right) wave buoys as well as satellite altimeters over the model domain (lowest). Scatter-qq plots for individual measurement sites can be found in Appendix X.1. The validation results of the multiyear product show good accuracy and are of the same order of magnitude as in other wave hindcasts produced for research purposes in the Baltic Sea (e.g., Tuomi et al., 2011; Björkqvist et al., 2018).

The overall accuracy of the significant wave height is good with combined bias of 0 m calculated over both open sea and coastal measurement sites and -0.02 m bias from altimeter (Table 1, Table 2 and Table 3). Difference of the bias between altimeters and wave buoys can partly be explained by the slight overestimation of SWH in most altimeters in CMEMS products (see https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-WAV-QUID-014-001.pdf). Even though combined bias is small, there is large variability between measurement sites, as bias at different sites varies from -0.05 m to 0.11 m at the open sea areas, between -0.15m up to 0.16 m by the coastal measurement sites (Table 7), and between -0.06 and 0.04 m when compared to altimeter measurements (Table 8).

The overall RMSD at open sea measurement sites was 0.21 m and had a very similar value at coastal sites (0.22 m). This varied largely between different measurement sites as RMDS varied from 0.18 m by Bothnian Bay up to 0.35 m by Hoburg1. Overall RMSD was largest along altimeter tracks (0.24 m) but here variation between altimeters was smaller than with buoys (0.22-0.27 m). Correlation between model and measurements was high and ranged up to 0.97 at Northen Baltic and Sodra Ostersjon as well as in nearby coastal Vilsandi measurement station. Lowest correlation of 0.89 was measured at Suomenlinna station, which locates inside Helsinki archipelago region and is already in an area that is at the limit of the capabilities of the model with 1 nmi resolution. Differences at quality metrics between different buoy sites are partly due to different measurement devices used and largely varying measurement periods and locations. Individual scatter-qq plots (Appendix X.1) highlight the differences at different sites. The scatter-qq plots in Appendix X.1 also show that the system has a tendency to slightly underestimate the significant wave height in open sea areas (e.g., NorthernBaltic, SodraOstresjon). Closer to coastal areas, the significant wave height is slightly overestimated (e.g., at Vilsandi outer and IsokariEast). The resolution of the ERA5 wind fields is relatively coarse, 0.28 degrees (approximately 31 km), considering the small size and shape of the Baltic Sea and the resolution of the BAL MFC wave model (ca. 1 nmi). Although the underestimation seen in the ERA5 wind speeds has been somewhat compensated by increasing the wind-wave coupling Betamax-parameter, it still affects the results. Effects on accuracy near coastal areas are difficult to estimate due to the limited number of observations in these areas; as data from many coastal measurements have been collected during measurement campaigns, they cover only short periods at maximum up to few years. Also, resolution and update frequency of the ice information used increased within hindcast period.

The hindcast is able to reproduce well the extreme wave events in the Northern Baltic Sea in December 1999, December 2004, and January 2005 (time series shown in Appendix X.4.1). The long model period

has also year-to-year variability in the results. Appendix X.4.2 shows scatter-qq plots and time series for two separate years (2005 and 2015) at Helsinki Buoy. In 2005 the model results underestimate the significant wave height but in 2015 the model shows better accuracy. Due to the complex shape of the Baltic Sea region, accurate wind fields are necessary for the wave model to perform well. The amount of data used for assimilation of the ERA5 reanalysis has increased over the years, which is expected to affect the accuracy of the forcing ERA5 wind fields and lead to better accuracy towards the end of the hindcast period.

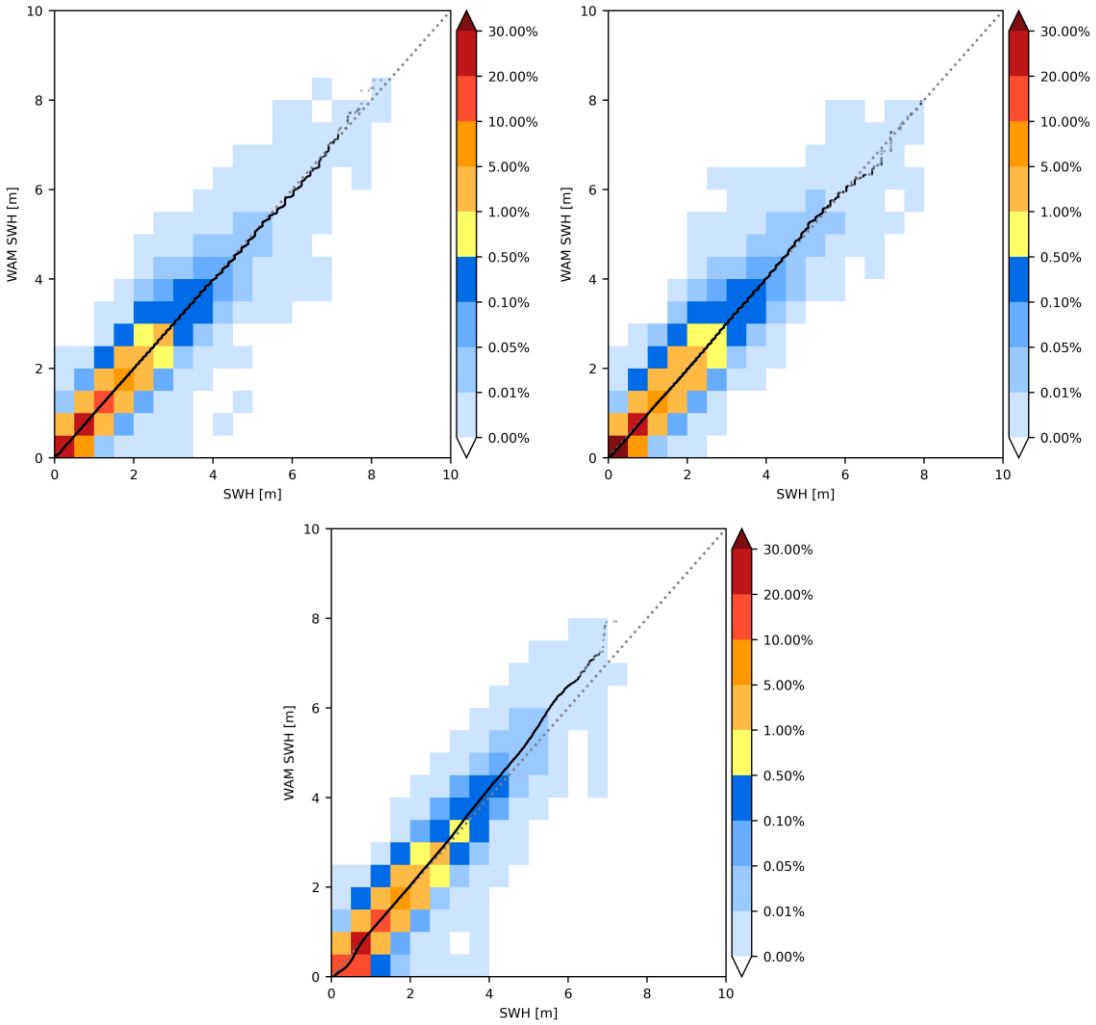

Figure 3 *Scatter-qq plot of significant wave height comparing hindcast significant wave heights to observations from 12 open sea Baltic Sea wave measurement sites listed in Table 4 (upper left panel), 12 coastal measurement sites listen Table 5 (upper right pane*

Table 7 *Statistical quantities of significant wave height at the 24 wave measurement sites. First 12 open sea stations followed by 12 coastal ones starting from Suomenlinna.*

Buoy Name	Mean obs	Mean model	Std obs	Std model	Bias	MAD	RMSD	Corr	SI	N
Olands Sodra Grund	1.03	1.14	0.7	0.79	0.11	0.19	0.28	0.95	0.27	135,143
Ostergarn	0.83	0.92	0.52	0.58	0.09	0.16	0.22	0.93	0.27	4,863
DarsserSWR	0.77	0.75	0.49	0.51	-0.03	0.14	0.19	0.93	0.25	145,561
Northern Baltic	1.25	1.20	0.9	0.87	-0.05	0.16	0.22	0.97	0.17	206,893
Laso Ost	0.81	0.81	0.5	0.61	0.00	0.16	0.21	0.95	0.26	32,914
Huvudskar Ost	0.94	0.97	0.65	0.69	0.03	0.14	0.20	0.96	0.21	83,888
Sodra Ostersjon	1.23	1.22	0.89	0.87	-0.01	0.17	0.23	0.97	0.18	32,078
FinngrundetWR	0.85	0.87	0.62	0.64	0.02	0.15	0.20	0.95	0.24	132,865
Bothnian Sea	1.04	1.03	0.78	0.76	-0.01	0.16	0.21	0.96	0.20	58, 883
Bothnian Bay	0.80	0.81	0.61	0.61	0.00	0.13	0.19	0.95	0.23	30,788
Helsinki buoy	0.87	0.81	0.60	0.60	-0.06	0.14	0.19	0.96	0.21	143,953
Knollsgrund	0.96	1.01	0.64	0.7	0.05	0.17	0.23	0.95	0.24	93,862
Suomenlinna	0.34	0.34	0.25	0.26	0.00	0.09	0.12	0.89	0.35	40,865
Rauma Outter	0.71	0.74	0.61	0.65	0.04	0.13	0.20	0.96	0.28	7,494
Gustav Dahlen	0.68	0.74	0.52	0.55	0.06	0.13	0.19	0.95	0.28	24,603
Laholmsbukten	0.58	0.5	0.49	0.44	-0.08	0.13	0.19	0.94	0.32	8,839
Gotland	0.91	0.86	0.62	0.59	-0.05	0.14	0.19	0.96	0.21	111,283
Ulkokalla	0.74	0.59	0.51	0.47	-0.15	0.19	0.25	0.92	0.34	1,625
Hoburg1	0.91	1.08	0.56	0.78	0.16	0.21	0.35	0.94	0.38	5,008
Vilsandi Outer	1.14	1.14	0.83	0.86	0.00	0.14	0.21	0.97	0.18	6,135
Trubaduren	0.67	0.71	0.57	0.64	0.04	0.15	0.22	0.94	0.32	9,623
Vaderoarna WR	1.11	1.08	0.85	0.87	-0.04	0.17	0.24	0.96	0.22	126,385
Almagrundet	0.84	1.01	0.66	0.74	0.16	0.21	0.29	0.95	0.35	54,246
Isokari East	0.73	0.81	0.61	0.67	0.08	0.18	0.25	0.94	0.34	5,086

Table 8 *Statistical quantities of significant wave height measured by 12 satellite altimeters.*

Satellite Altimeter	Mean obs	Mean model	Std obs	Std model	Bias	MAD	RMSD	Corr	SI	N
Envisat	1.06	1.03	0.7	0.76	-0.03	0.17	0.23	0.95	0.22	182,564
Jason-1	1.10	1.05	0.67	0.76	-0.06	0.19	0.25	0.95	0.23	306,184
Jason-2	1.02	1.02	0.68	0.73	-0.01	0.17	0.23	0.95	0.23	293,889
Jason-3	1.04	1.02	0.68	0.74	-0.01	0.19	0.25	0.94	0.24	209,874
SARAL/Altika	1.06	1.04	0.70	0.73	-0.02	0.17	0.22	0.95	0.21	199,094
Sentinel-3A	1.04	1.00	0.61	0.69	-0.04	0.2	0.26	0.93	0.25	69,903
Sentinel-3B	1.07	1.03	0.68	0.76	-0.04	0.21	0.27	0.94	0.25	67,635
Sentinel-6A	1.28	1.32	0.67	0.72	0.04	0.18	0.25	0.94	0.19	47,511
HaiYang-2B	1.13	1.15	0.67	0.73	0.02	0.17	0.23	0.95	0.21	45,926
HaiYang-2C	1.15	1.16	0.71	0.78	0.00	0.18	0.24	0.95	0.21	31,146
Cryosat-2	1.04	1.05	0.67	0.74	0.01	0.18	0.24	0.95	0.23	70,635
SWOT nadir	1.11	1.09	0.75	0.81	-0.02	0.19	0.26	0.95	0.23	9,320

IV.2 Wave period at spectral peak

The hindcast wave period at spectral peak was compared to observations from 15 Baltic Sea measurement sites. The statistical quantities for each measurement site are presented in Table 9. The scatter plots (Figure 4) illustrate the wave model quality over the open sea areas and coastal stations (Table 7). Scatter plots for individual measurement sites can be found in Appendix 2.

The statistical results of wave period at spectral peak in Table 9 are typical for wave models (e.g., Tuomi, 2008). The bias is negative at all measurement sites indicating that the model slightly underestimates the peak period. Compared to significant wave height, the scatter in the wave peak period is much larger. Mostly correlation varies between 0.60 and 0.91 and RMSD between 0.64 s (Bothnian Sea) and 1.32 s (Helsinki Buoy). The representation of peak period is very sensitive on the wave model's ability to accurately simulate the wave energy for the different components of the wave spectrum (wind sea, swell). Thus, the modelled value may represent the peak period of the wind sea whereas the measured value represents that of the swell, or vice versa. Pettersson et al. (2010) have shown that WAM is not able to reproduce the steering effect of the narrow gulf on the wave field in the Gulf of Finland. This could explain the lower correlation (0.65) and larger bias (-0.33 s) and RMSD (1.32 s) in the peak period at Helsinki buoy. High RMSD (1.76 s) by VarderoarnaWR is estimated to result from the close proximity to the model boundary. Comparison to the Suomenlinna buoy (correlation of 0.24 and RMSD of 1.95 m, Appendix Figure 10) shows how 1 nmi model has challenges in representing peak period inside archipelago. This can also be explained by different nature of the wave spectrum inside archipelago where this has been shown to have almost constant energy with wide range of frequencies (Björkqvist et al. 2019). This spectral shape makes estimation of the peak value, challenging both from the measured and modelled wave fields. Validation of the SWH at the same location shows that model estimates energy of the wave fields adequately, considering models resolution (seeTable 7 and Appendix Figure 7).

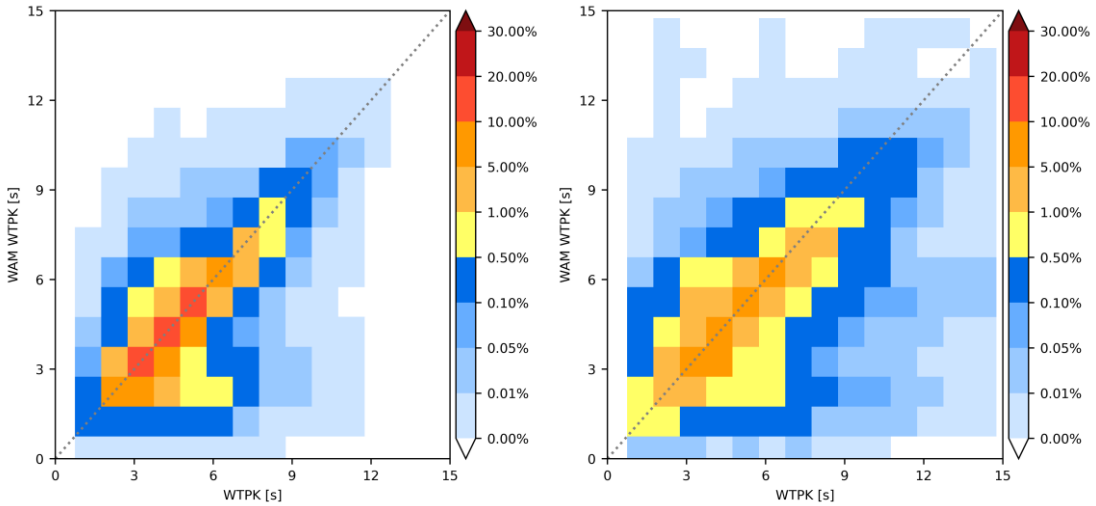

Figure 4 Scatter plot comparing hindcast spectral peak periods to the measurements t 9 open sea measurement sites (left panel, measurement sites listed in Table 4 and by 6 coastal stations (right panel, measurement sites listed in Table 5). The colours represent the percentage of the total number of entries. Scatterplots for individual measurement sites can be found in Appendix 2.

Table 9 *Statistical quantities of wave period at spectral peak at 15 wave measurement sites. First 9 open sea stations followed by 6 coastal ones starting from Suomenlinna.*

Buoy Name	Mean obs	Mean model	Std obs	Std model	Bias	MAD	RMSD	Corr	SI	N
DarsserSWR	4.35	4.08	1.13	1.22	-0.26	0.72	1.09	0.60	0.25	99,122
Northern Baltic	5.43	5.16	1.59	1.62	-0.26	0.61	0.93	0.85	0.17	206,895
HuvudskarOst	4.94	4.82	1.35	1.32	-0.11	0.59	0.89	0.78	0.18	46,890
Sodra Ostersjon	5.45	5.18	1.61	1.55	-0.26	0.58	0.85	0.87	0.16	31,567
FinngrundetWR	4.4	4.27	1.35	1.33	-0.14	0.54	0.77	0.84	0.17	130,642
Bothnian Sea	4.68	4.48	1.42	1.36	-0.21	0.48	0.64	0.91	0.14	58,883
Bothnian Bay	4.17	4.00	1.27	1.30	-0.17	0.48	0.67	0.87	0.16	30,788
Helsinki Buoy	4.87	4.55	1.49	1.55	-0.33	0.84	1.32	0.65	0.27	143,953
Knollsgrund	4.96	4.92	1.47	1.44	-0.04	0.72	1.10	0.71	0.22	83,554
Suomenlinna	3.95	3.68	1.59	1.55	-0.26	1.27	1.94	0.25	0.49	40,865
Rauma Outter	4.12	4.06	1.32	1.27	-0.06	0.48	0.68	0.86	0.17	7,494
Goltand	5.22	5.09	1.48	1.52	-0.13	0.65	0.96	0.89	0.18	111,283
Ulkokalla	4.05	3.93	1.26	1.36	-0.13	0.62	0.89	0.78	0.22	1,621
VaderoarnaWR	5.90	5.47	2.25	1.92	-0.43	1.11	1.76	0.67	0.30	124,665
Isokari East	4.20	4.15	1.62	1.48	-0.06	0.63	0.81	0.87	0.19	5,086

IV.3 Mean wave period

The hindcast mean wave period was compared to observations from 15 Baltic Sea measurement sites (8 open sea sites). The statistical quantities for each measurement site are presented in Table 10. The scatter plot in Figure5 shows the wave model quality over the 8 open sea measurement sites. Scatter plots for individual measurement sites can be found in Appendix 3.

The results in Table 10 show that the mean wave period is well reproduced by the model. Correlation varies between 0.65 and 0.95, which is slightly better than for the wave period at spectral peak. The bias is negative at all buoy locations. The scatter plots in Figure 5 shows the systematic underestimation of the mean wave period and how the underestimation and scatter is larger at coastal stations; especially by buoys in the Skagerrak region (VarderoarnaWR and Trubaduren, see Appendix Figure 12).

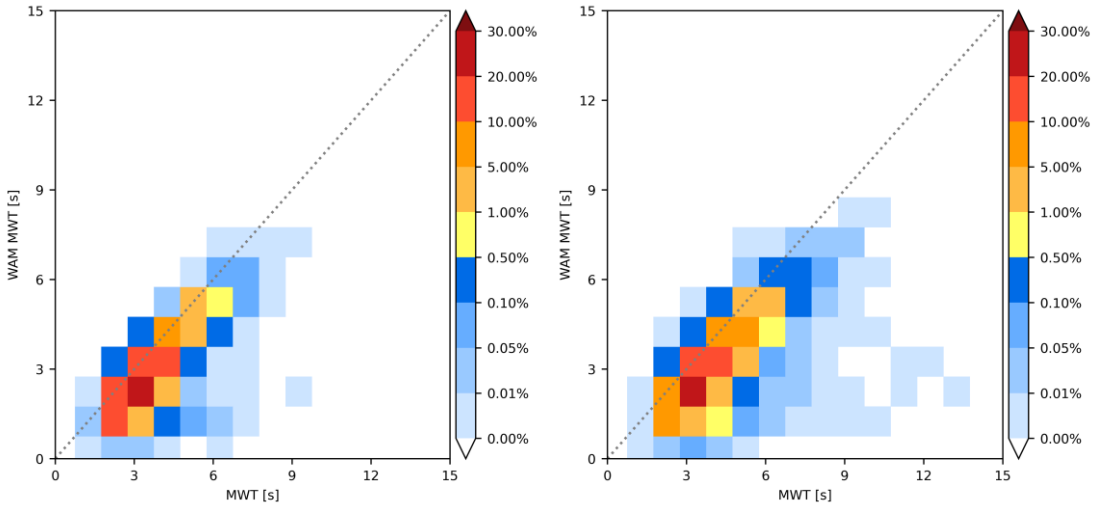

Figure 5 *Scatter plot comparing hindcast mean wave period to the measurements at 8 open sea measurement sites (left panel, measurement sites listed in Table 4) and by 7 coastal stations (right panel, measurement sites listed in Table 5). The colours represent the percentage of the total number of entries. Scatterplots for individual measurement sites can be found in Appendix 3.*

Table 10 *Statistical quantities of mean wave period at 15 wave measurement sites.*

Buoy Name	Mean obs	Mean model	Std obs	Std model	Bias	MAD	RMSD	Corr	SI	N
Olands Sodra Grund	4.03	3.26	0.93	0.93	-0.77	0.77	0.86	0.91	0.21	135,143
Ostergarn	3.69	3.05	0.77	0.78	-0.63	0.65	0.74	0.87	0.20	4,708
DarsserSWR	3.34	2.55	0.61	0.71	-0.79	0.79	0.97	0.65	0.29	125,170
LasoOst	3.35	2.60	0.51	0.78	-0.75	0.76	0.89	0.81	0.27	32,460
HuvudskarOst	3.68	3.03	0.74	0.86	-0.65	0.66	0.74	0.91	0.2	33,762
Sodra Ostersjon	3.96	3.37	0.97	1.00	-0.59	0.61	0.68	0.94	0.17	31,563
FinngrundetWR	3.40	2.81	0.74	0.82	-0.59	0.6	0.67	0.92	0.20	132,841
Knollsgrund	3.60	3.04	0.77	0.84	-0.56	0.58	0.68	0.89	0.19	91,715
Gustav Dahlen	3.67	2.78	0.86	0.86	-0.89	0.89	1.01	0.85	0.28	23,962
Laholmsbukten	3.11	2.31	0.77	0.88	-0.79	0.8	0.94	0.82	0.30	8,626
Hoburg1	3.83	3.19	0.82	0.91	-0.64	0.65	0.73	0.92	0.19	4,743
Vilsandi Outer	3.81	3.30	0.93	1.05	-0.52	0.53	0.61	0.95	0.16	6,008
Trubaduren	3.76	2.64	0.73	0.95	-1.13	1.13	1.30	0.74	0.34	9,623
VaderoarnaWR	4.02	3.28	1.07	1.12	-0.74	0.76	0.93	0.87	0.23	126,237
Almagrundet	3.91	3.05	0.94	0.92	-0.86	0.86	0.97	0.88	0.25	50,791

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This is a new version of the hindcast. The model version was updated to 4.7 and source term package was changed to ST4. Due to change of source term package, Betamax was re-tuned to match ERA5 wind fields. Bathymetry and obstruction fields were updated based on EMODnet bathymetry to improve results especially in the coastal areas.

Multiple new variables were also introduced to the hindcast. Two variables for the highest waves:

- Height of the highest crest
- Maximum crest trough wave height (Hc,max)

and new dataset with following sea-air fluxes:

- Wave Momentum Flux into Sea Water
- Surface Downward Eastward (and Northward) Stress due to Ocean Viscous Dissipation
- Charnock coefficient
- Surface Roughness Length

The product was also extended back in time to cover years 1980-1992. Monthly climatology was recalculated with the new hindcast version.

VI QUALITY CHANGES SINCE PREVIOUS VERSION

The new hindcast has been produced using an updated version of WAM (4.7) and new source term package for parametrization for dissipation of wind-generated waves (ST4-physics, Ardhuin et al., 2010). This source term package has been shown to give best results out of the available source term packages in the Baltic Sea (Björkqvist et al., 2020) especially by the coastal region. Improvements by the coastal region were enhanced by the update of the bathymetry and the obstruction fields based on the open data available through EMODnet bathymetry.

In this hindcast, the improvements in the SWH at all the measurements stations is seen with slightly decreased RMSD and increased correlation. Tuning of Betamax parameter for ERA5 winds has also helped to reduce slight underestimation of the SWH from the previous version of the product, where it was -0.04 m over the whole model domain. Most notable improvement of the product quality is seen in the prediction of spectral peak period (WTPK, Chapter IV.2). Bias of the previous version has been decreased by up to half from the old version and there is also a slight systematic decrease in RMSD and improved correlation at all stations previously validated by the hindcast.

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VIII APPENDIX

VIII.1 SCATTER PLOTS OF SIGNIFICANT WAVE HEIGHT

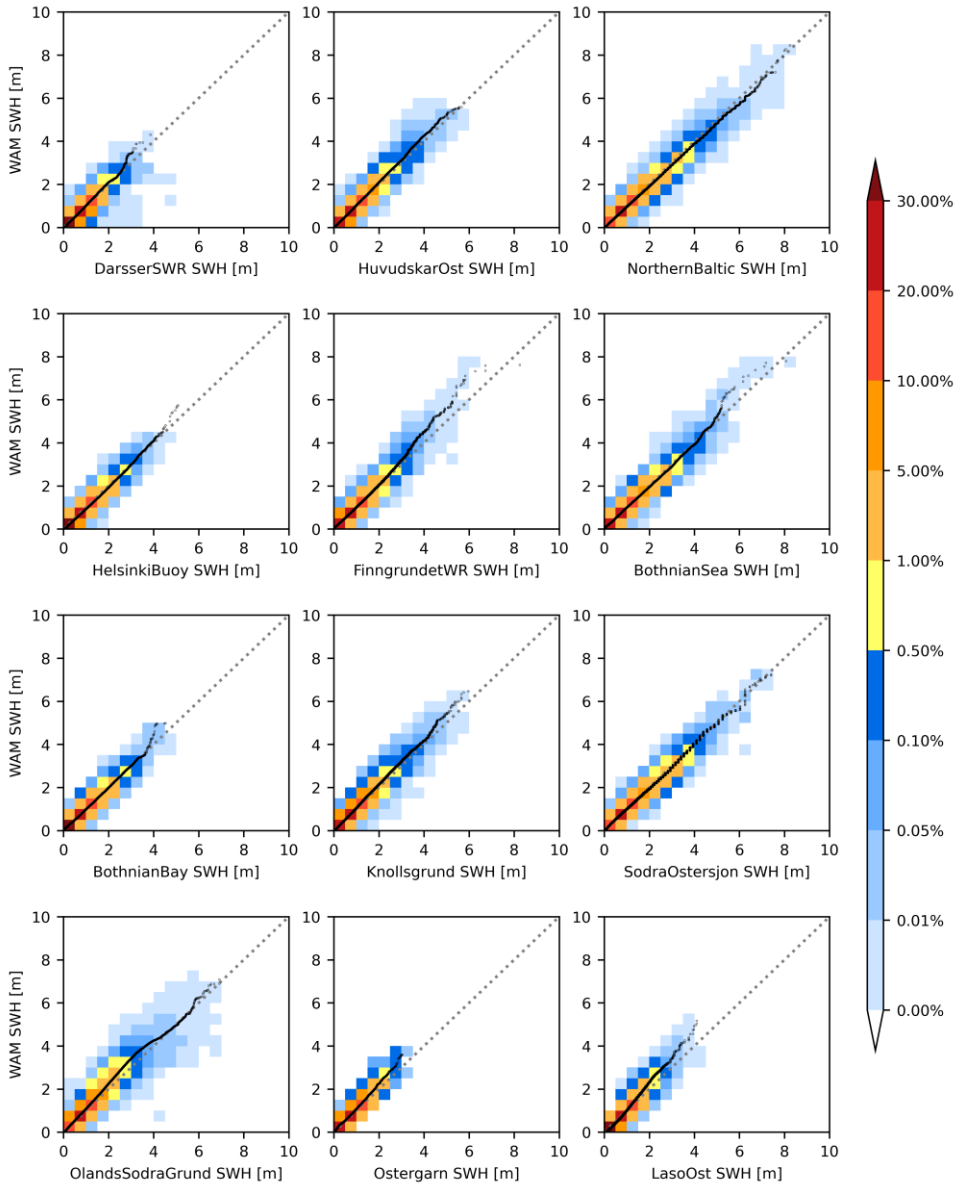

Figure 6 *Scatter plots of significant wave height from 12 individual measurement sites representing open sea measurement sites listed in Table 4. The colours represent the percentage of the total number of entries.*

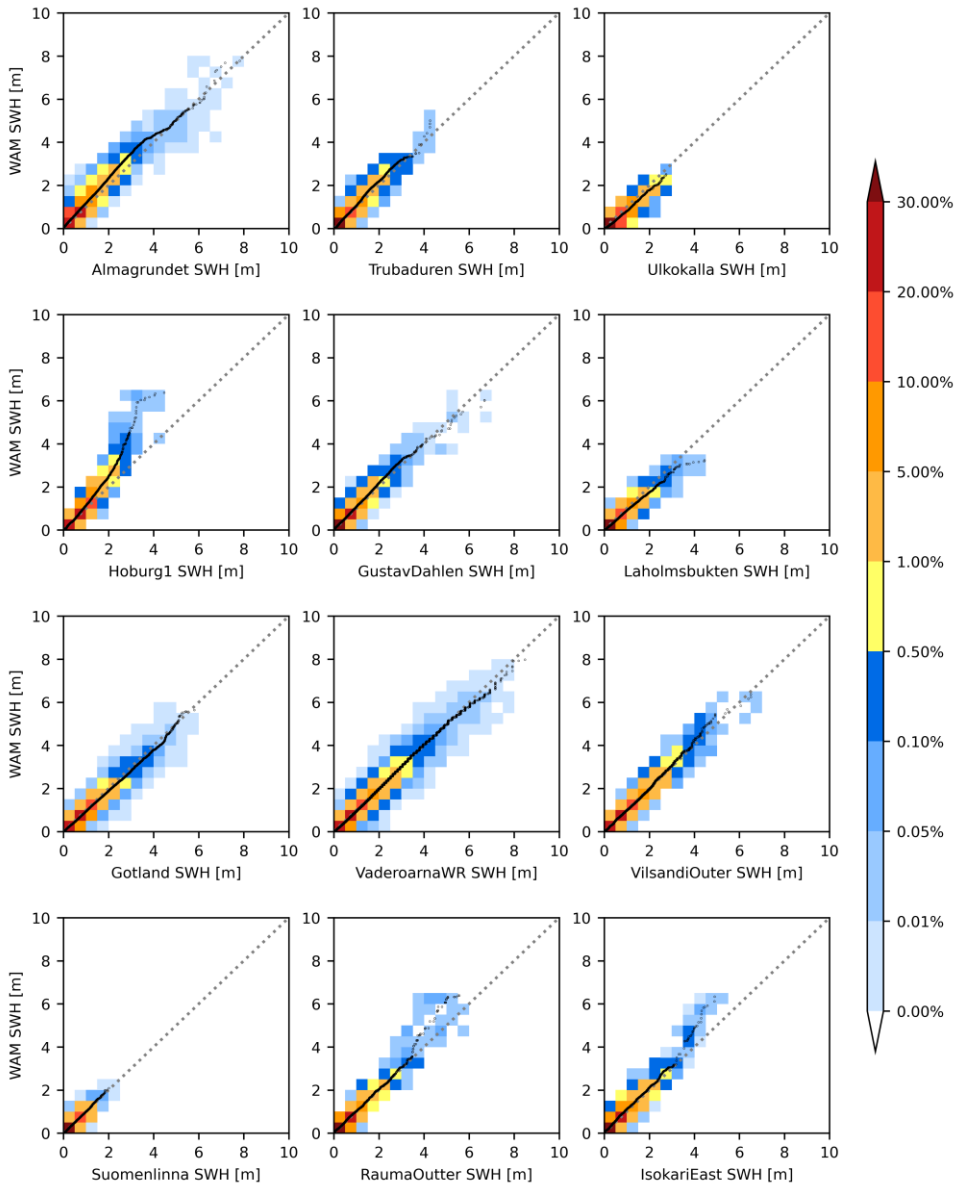

Figure 7 *Scatter plots of significant wave height from 12 individual measurement sites representing coastal measurement sites listed in Table 5. The colours represent the percentage of the total number of entries.*

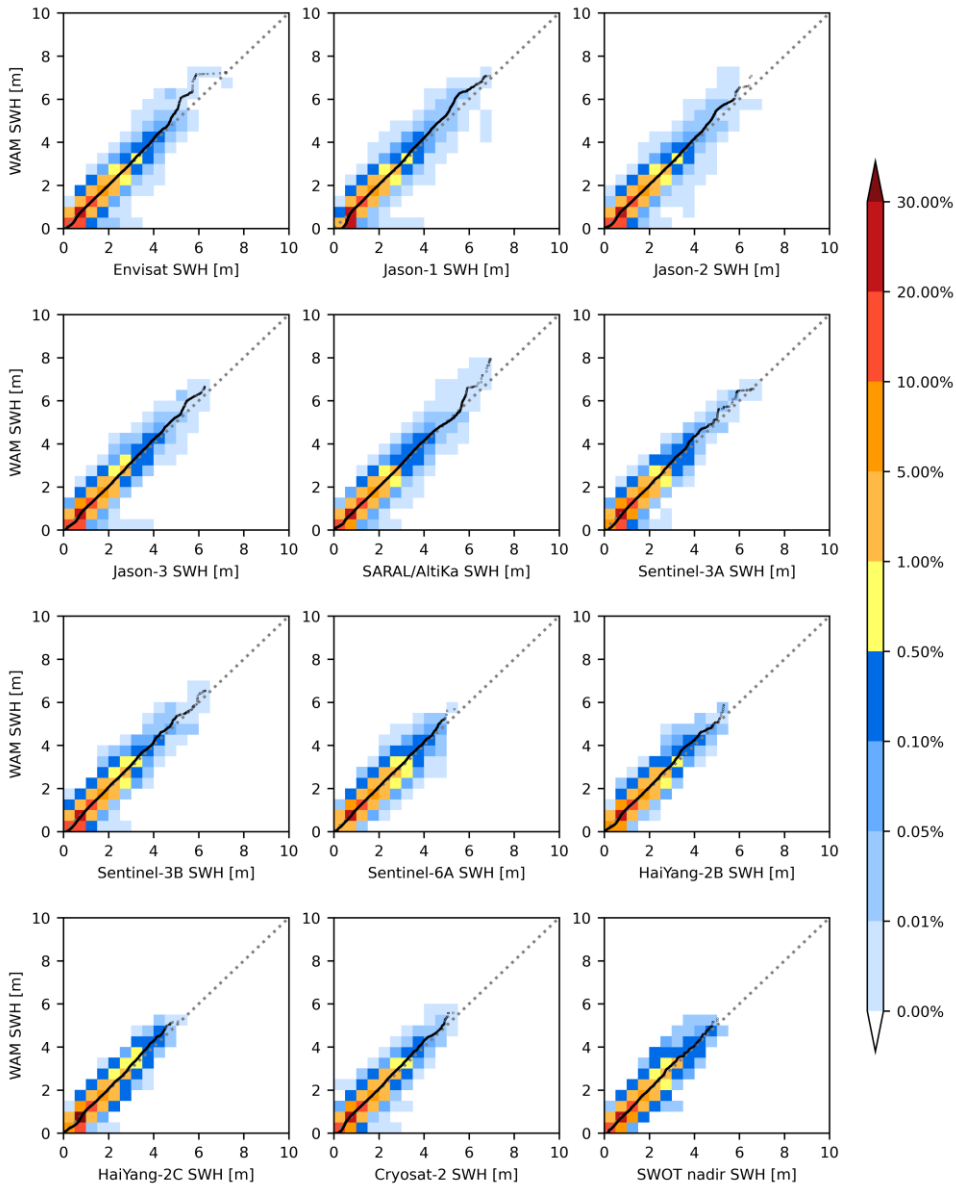

Figure 8 *Scatter plots of significant wave height from 12 individual satellite altimeters listed in Table 6. The colours represent the percentage of the total number of entries.*

VIII.2 SCATTER PLOTS OF SIGNIFICANT PEAK PERIOD

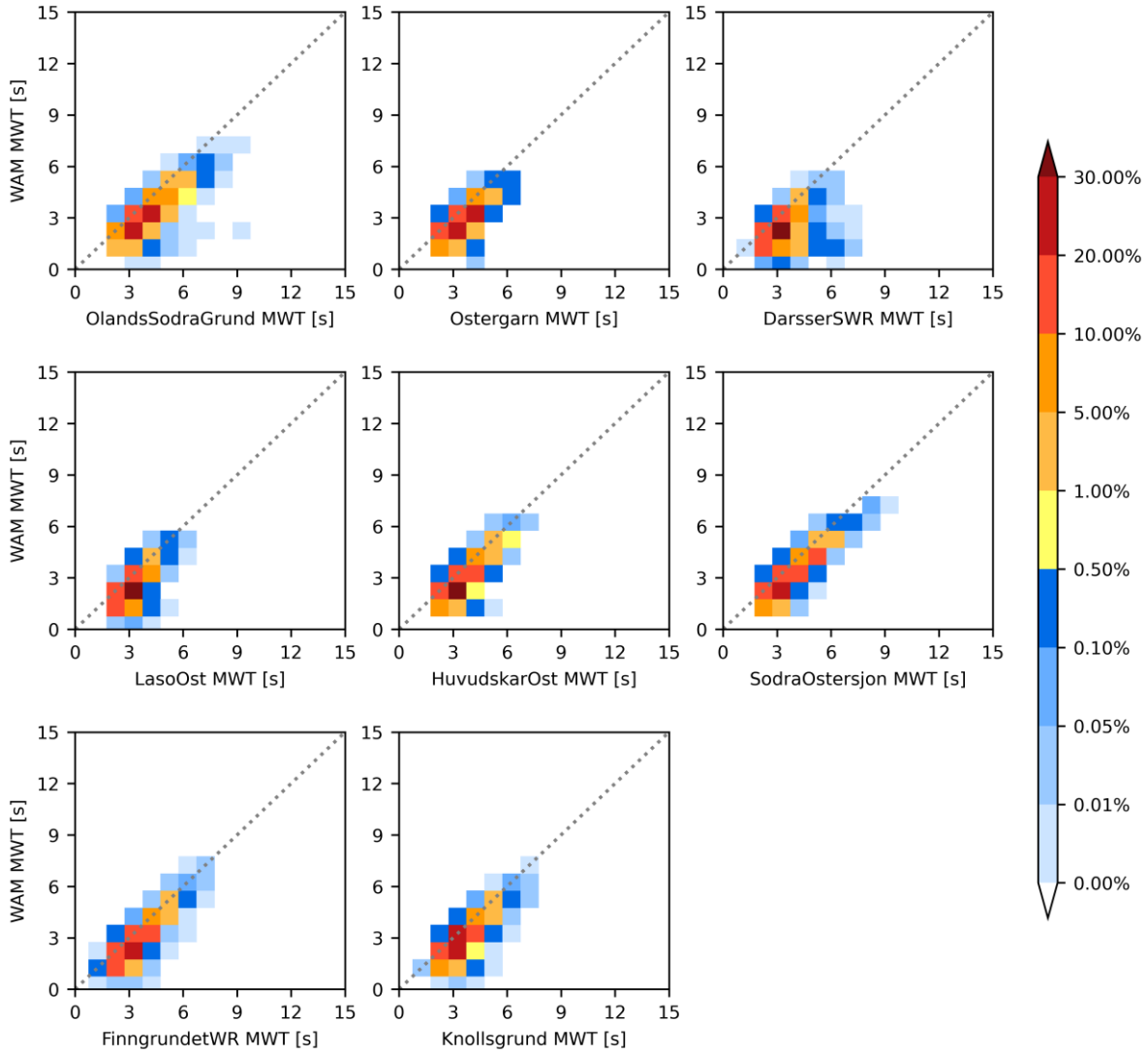

Figure 9 *Scatter plots of wave period at spectral peak (Tp) from 12 individual measurement sites representing open sea measurement sites listed in Table 4. The colours represent the percentage of the total number of entries.*

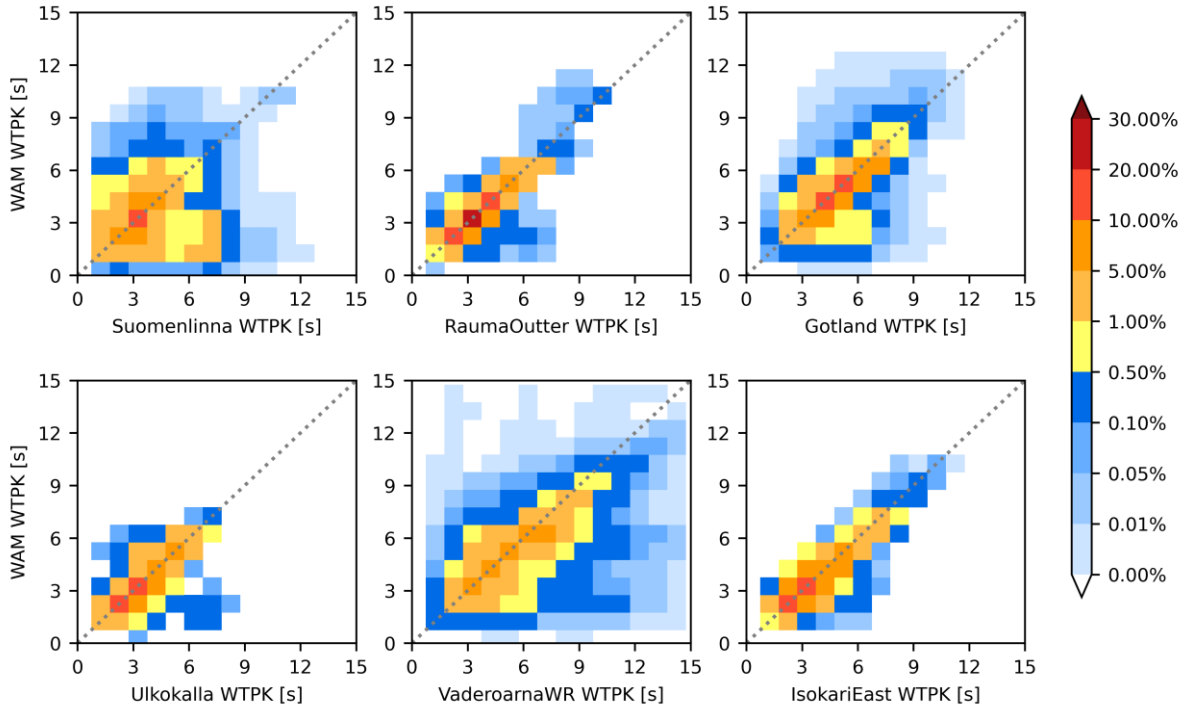

Figure 10 *Scatter plots of wave period at spectral peak (Tp) from 12 individual measurement sites representing coastal measurement sites listed in Table 5. The colours represent the percentage of the total number of entries.*

VIII.3 SCATTER PLOTS OF MEAN PERIOD

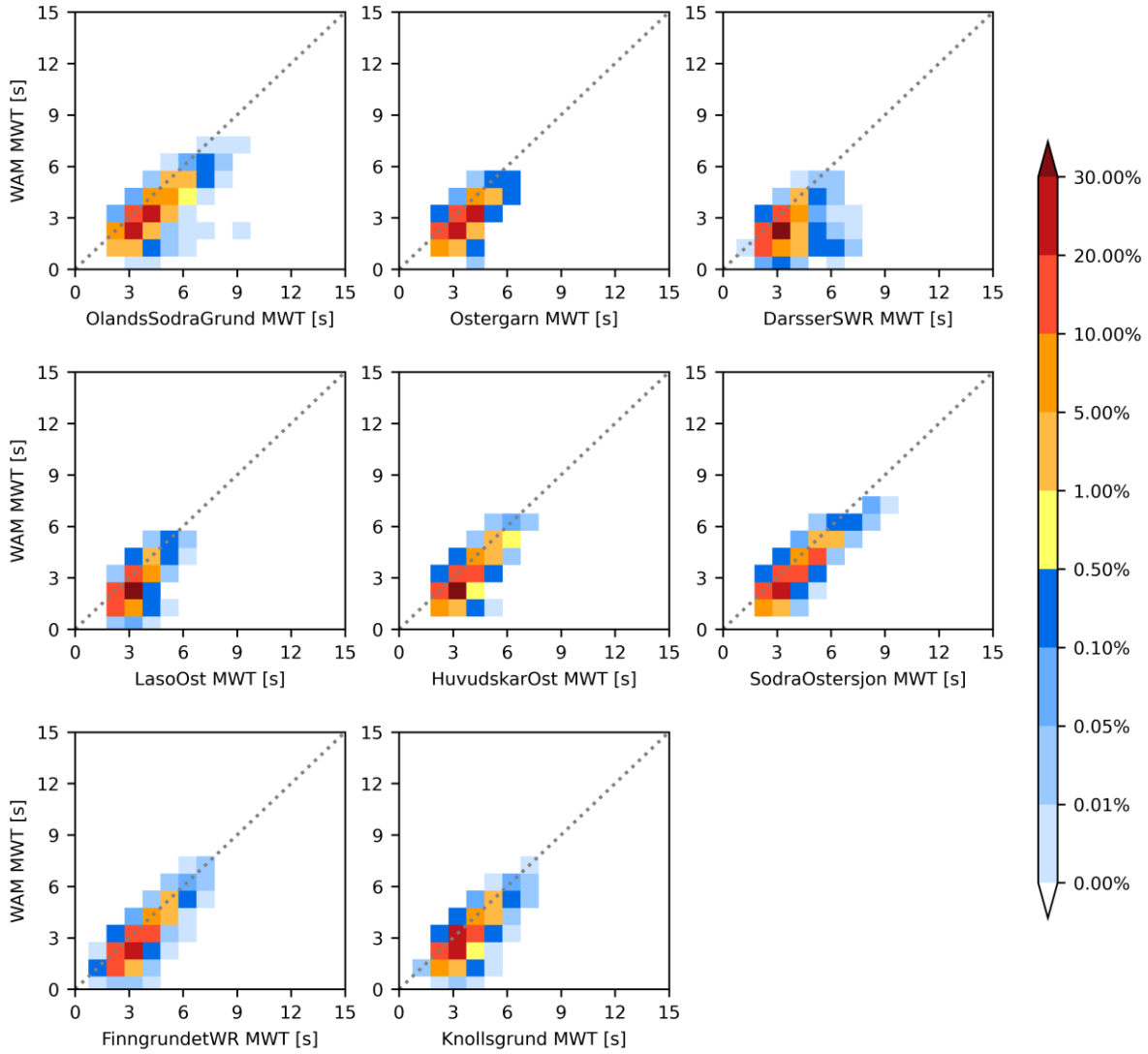

Figure 11 *Scatter plots of average zero crossing period (Tm) from 8 individual measurement sites representing open sea measurement sites listed in Table 4. The colours represent the percentage of the total number of entries.*

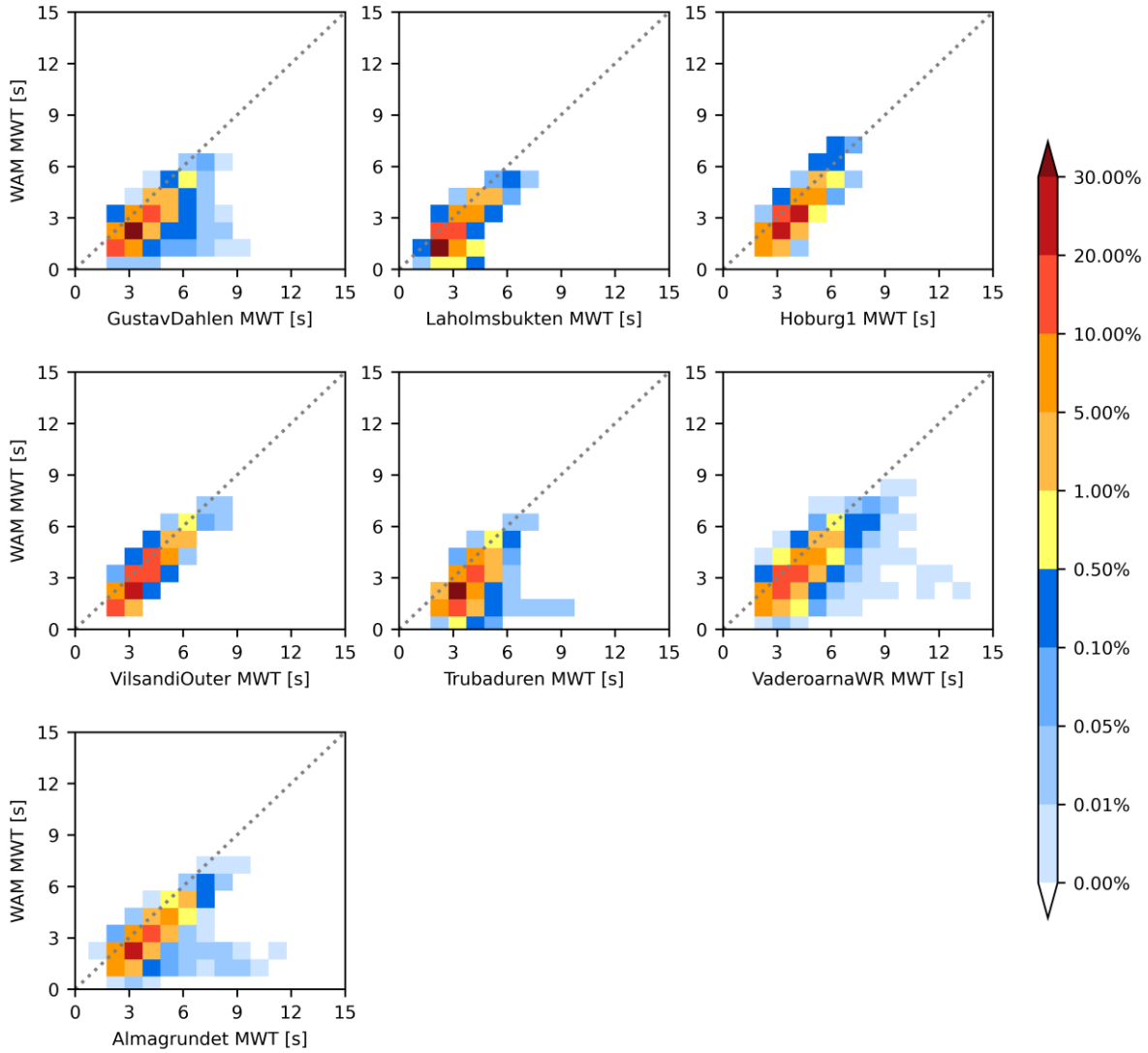

Figure 12 *Scatter plots of average zero crossing period (Tm) from 12 individual measurement sites representing coastal measurement sites listed in Table 5. The colours represent the percentage of the total number of entries.*

VIII.4 TIME SERIES

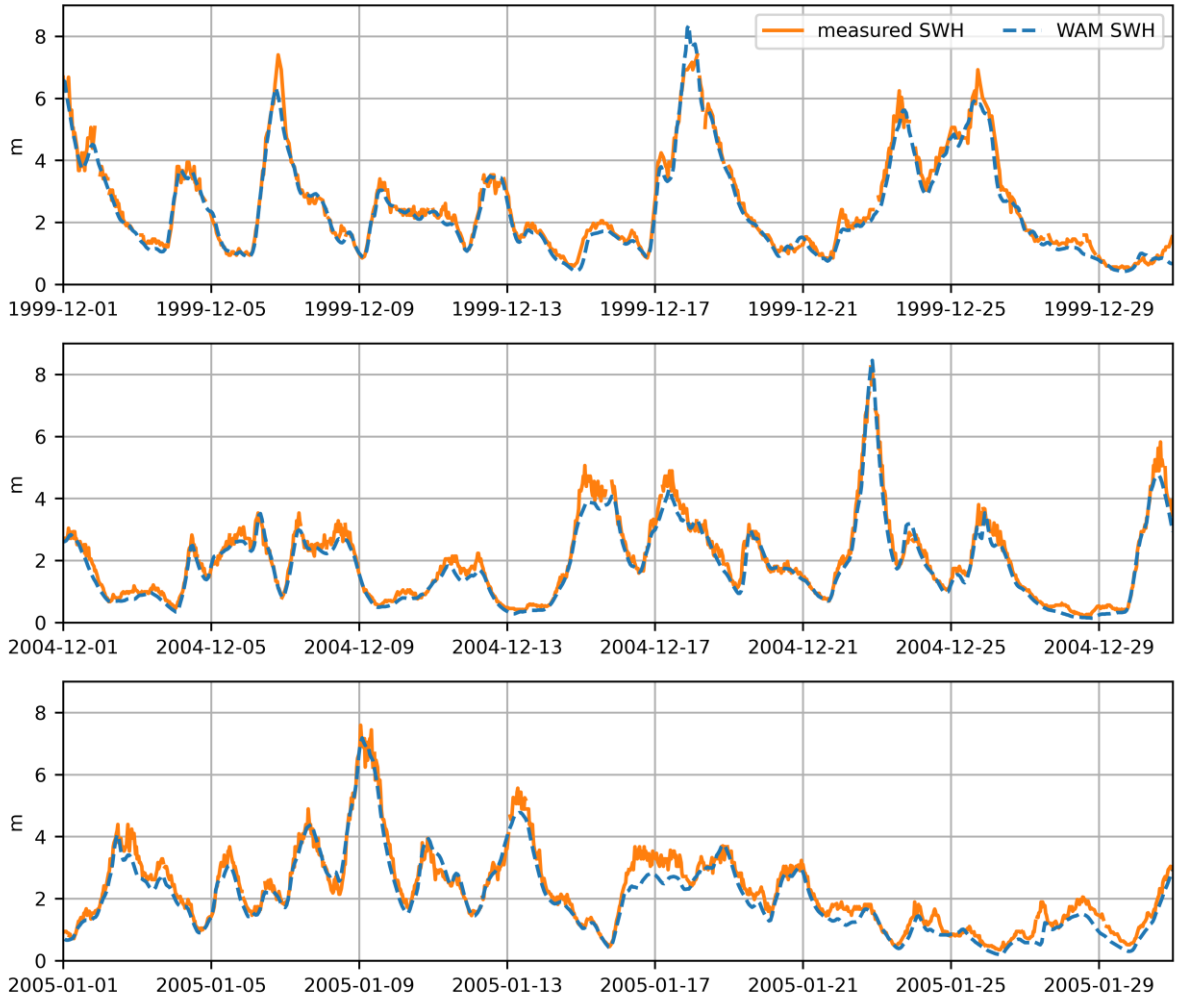

Figure 13 *Time series of significant wave height (SWH) from the Northern Baltic buoy in December 1999 (upper), December 2004 (middle), and January 2005 (lower). The orange line represents buoy measurements and the blue dashed line hindcast result*

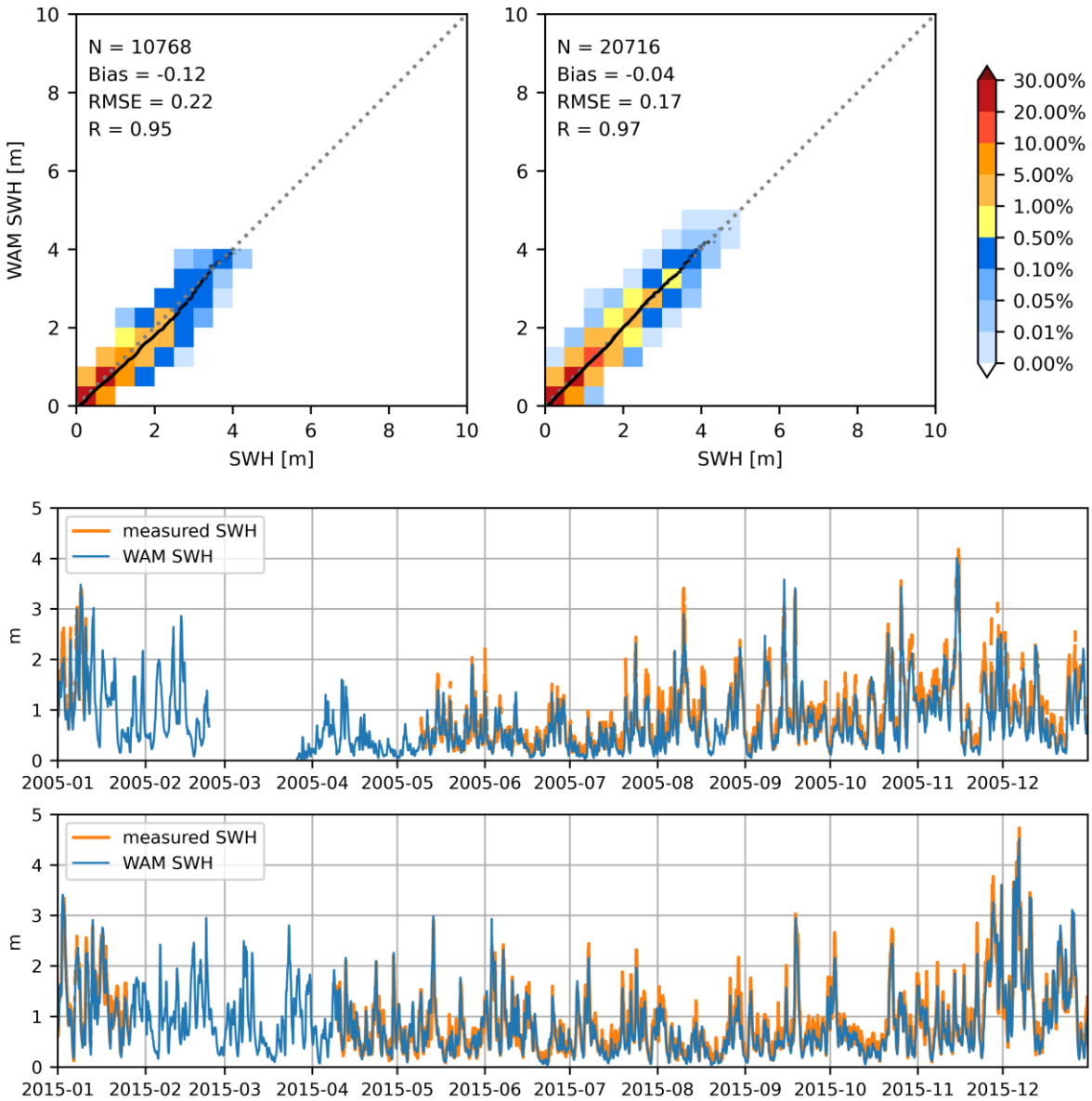

Figure 14 *Scatter-qq plots (upper panels) and time series (middle and bottom panel) of significant wave height at Helsinki Buoy in 2005 and in 2015. Left scatter-qq plot and upper timeseries represent year 2005 and right scatter-qq with bottom timeseries year 2015. Scatter colours represent the percentage of the total number of entries. In the time series, the orange line shows buoy measurements and the blue wave model results.*